START HERE

Football tracking, explained simply

Every feature.
Every calculation.
Every validation limit.

A complete guide to the Possession Demo

Prepared for the tracking and events analysis of match 3816: Argentina vs Saudi Arabia. 2 November 2026.

What this guide gives you

Simple meanings, step-by-step rules, made-up examples, real benchmark counts, the app controls, exported fields and what still needs validation.

The most important distinction

A touch, a player controlling the ball, and a team retaining possession are different things. The current program estimates them separately.

The learned control classifier is untrained. Provisional possession is not an official match statistic. All event labels are candidates unless the stated validation supports them.

Main result: completed passes on smoothed tracking - 79.4% precision and 84.3% recall.

Code and saved results were inspected for this guide. Examples labelled "Example" are illustrations, not extra measurements. See the final pages for sources and technical formulas.

START HERE

How to read this guide

You do not need to read code or know machine learning. Start with the meanings, then read each feature like a small recipe.

Word in this guide	Very simple meaning
Observed	A coordinate or label was supplied in a file. It can still contain error.
Estimated	The program worked out a likely answer from movement.
Candidate	The pattern looks like an event. It is not a confirmed official decision.
Validated	We compared a prediction with a particular set of reference labels.
Unknown / unassigned	The program did not supply an answer. This is not proof that nobody had the ball.
Not measured	We do not have an appropriate accuracy result for this target.

The order of the guide

Inputs and movement come first. Player interaction, control and possession follow. Then every event rule is explained. Validation, results, feature columns, app controls and technical details come last.

Read the denominator

93.1% touch-reference recovery, 97.5% agreement when a carrier was assigned, and 44.4% team assignment coverage answer three different questions. They must not be read as the same accuracy.

Every page has a bookmark in the PDF outline. Use your PDF viewer's bookmarks panel to jump to a feature.

FIND A FEATURE

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FIND A FEATURE

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Use the page numbers below or your PDF viewer's bookmarks. Calculations and validation are explained separately so each score keeps its correct meaning.

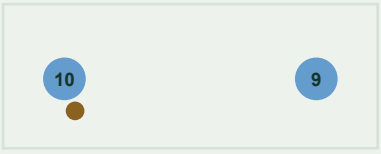
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FOUNDATIONS

Touch, control and possession

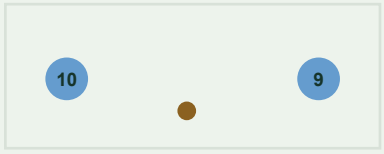
Concept	Meaning	Example
Touch / kick	The player may have changed the ball's motion.	A one-touch pass.
Interaction	The player is in a position to act on the ball.	A player receives and then releases it.
Likely control	Movement fits our simple control rule.	A player runs with a nearby ball.
Team possession	The team is provisionally assigned ownership.	The team can retain possession while a pass travels.

1 Player 10 interacts



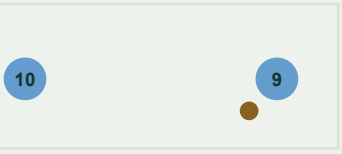
Current interaction: 10

2 Ball travels



No individual carrier

3 Player 9 interacts



Current interaction: 9

Why the distinction matters

Player 10 can touch a pass immediately without holding it for long. While the ball travels, no individual carrier may be assigned. Team possession can still be retained provisionally until another player interacts.

Current model versus football truth

The word carrier in the exported frame file means estimated physical interaction. It is not a directly observed foot contact and not validated sustained control.

INPUTS

What is in the input files?

The tracking file is JSONL: one JSON record on each line. Each record describes one frame of player and ball positions.

Input	What we use it for
frameNum	Identify the source frame.
videoTimeMs	Find the source video time; divide by 1,000 to get seconds.
period	Keep periods separate. Do not carry history over halftime.
Home/away player arrays	Team, jersey, x, y and tracking quality.
Ball fields	x, y, reported height z and visibility.
Optional events JSON array	Reference actors, outcomes and timestamps for validation after prediction.

Raw or smoothed?

Raw uses homePlayers, awayPlayers and balls. Smoothed uses homePlayersSmoothed, awayPlayersSmoothed and ballsSmoothed. Both options come from the same supplied tracking file.

Units and pitch

x and y must be centred pitch metres on a 105 m by 68 m pitch. The centre is (0, 0), goal lines are x = -52.5 and +52.5, and touchlines are y = -34 and +34. Pixel coordinates need conversion first.

Identity

home:10 and away:10 are different players. Array order can change; team plus jersey remains the identity. Names are joined afterwards from event roster information.

The loader supports up to 64 player identities across the capture. An empty or multiple-ball raw list does not cause an arbitrary ball selection; ball position is treated as unavailable.

INPUTS

The complete processing route

1. Read tracking positions and quality only. Resolve players by team plus jersey.
2. Remove repeated frames, handle short missing sections and quarantine implausible position changes.
3. Smooth movement and estimate velocities and ball acceleration.
4. Find possible touches and interaction starts. Build each player's interaction intervals.
5. Estimate the current carrier, heuristic likely control and provisional possession.
6. Build event candidates from geometry and changes of interacting player.
7. Freeze those predictions. Only now read the optional events file.
8. Match predictions to eligible references and export the results for the app.

No event-fed prediction

Embedded game_event, possession_event and their IDs are ignored by the predictor. The separate event file supplies validation and name lookups afterwards.

What was evaluated

The capture has 199,012 input records and 198,142 distinct retained frames. There are 870 repeated frame records. The smoothed run builds 1,708 interaction intervals and 7,207 overlapping candidate-event records.

What the counts do not mean

One real pass may create a touch, kick, completed-pass and reception record. Therefore 7,207 candidate records are not 7,207 separate confirmed football actions.

Implementation: [C1], [C2], [C5]. Benchmark: [B1].

PREPARATION

Cleaning duplicates and bad input

Duplicate rule

When adjacent records repeat the same frame number, source time and period, keep the first observation. Do not choose the version that best agrees with an event label.

Same frameNum + same videoTimeMs + same period -> retain first record

Example

Suppose one frame appears three times. Counting all three would distort event detection and frame totals. We keep one frame. If the repeated coordinates disagree, we count the conflict for the audit.

Audit item	Smoothed capture
Input records	199,012
Unique retained frames	198,142
Skipped duplicates	870
Conflicting selected smoothed duplicates	464
Ignored embedded game-event payloads	46,039
Ignored embedded possession-event payloads	2,114

Other input checks

Blank lines are ignored. Malformed JSON lines are skipped and counted. After duplicate removal, source times must be finite and strictly increasing. Structurally invalid records, such as missing required frame numbers, can stop the job with an error.

A repeat is not extra time

Playing-time calculations use differences between timestamps. Duplicate observations must not create extra possession seconds.

Implementation: [C1]. Counts compare duplicates with the retained first observation, not just with the immediately preceding duplicate.

PREPARATION

How tracking quality is used

Estimated positions can still contain useful information. The new model gives uncertain observations less influence during smoothing instead of rejecting every estimated coordinate.

Confidence label	Base weight
HIGH	1.00
MEDIUM	0.70
LOW	0.35
Missing / other label	0.60

Final weight = base confidence weight x visibility factor

Visibility	Factor
VISIBLE	1.00
Anything else, including estimated	0.50
Short reconstructed coordinate	Assigned a weight of 0.20 for smoothing

Examples

A HIGH, VISIBLE player has weight 1.00. A LOW, ESTIMATED player has 0.35 x 0.50 = 0.175. A visible ball with no confidence label normally has 0.60, because the default base weight is 0.60.

Weight is not accuracy

A weight of 0.60 does not mean a 60% chance of being correct. These are chosen smoothing weights, not calibrated probabilities.

The smoother makes low-weight measurements less precise. Its measurement-variance formula is given in the technical appendix.

PREPARATION

Missing positions and tracking jumps

Short missing sections

If valid coordinates exist before and after a small hole, the program can draw a straight interpolation between them. The observed endpoints must be no more than 0.3 seconds apart, in the same period, with no source-frame gap above 0.2 seconds.

Example

The ball is at x = 0 at 10.0 s and x = 2 at 10.2 s. If the 10.1 s coordinate is missing, the reconstructed x is 1. It is labelled reconstructed, not observed.

Reconstructed position = start + fraction of elapsed time x (end - start)

Check	Current default
Player transition speed	Above 15 m/s: quarantine the later coordinate.
Ball transition speed	Above 60 m/s: quarantine the later coordinate.
Filled ball frames in this run	6 smoothed; 47 raw.
Long missing ball sections	Remain unknown.

Order matters

Short holes are filled first. Implausible jumps are then quarantined. These newly removed jump samples are not automatically interpolated again. Smoothing resets across remaining missing sections, period changes and long source gaps.

What cannot be recovered

A long invisible ball flight can have many plausible paths. The model does not invent a complete trajectory or a carrier for it.

MOVEMENT

Movement: distance, speed and acceleration

Distance to the ball

Compare the player's x and y with the ball's x and y. A separation of 3 m across and 4 m along gives a distance of 5 m.

```
distance = square root of ((player_x - ball_x)^2 + (player_y - ball_y)^2)
```

Speed

Speed tells us how quickly the position changes. Moving 2 m in 0.5 s is 4 m/s. Multiplying m/s by 3.6 gives km/h. The displayed speed is the magnitude of the estimated xy velocity.

```
speed = square root of (velocity_x^2 + velocity_y^2)
```

Acceleration

Acceleration describes how velocity changes. We use the size of the xy acceleration vector to find possible contacts. A change in direction can produce acceleration even if speed stays similar.

Smoothing in simple words

The filter predicts the next position, compares it with the supplied observation, and blends them according to uncertainty. A backward pass also uses later observations to improve earlier estimates.

Offline processing

Because later frames are used, these are retrospective results. A smooth replay is not proof that the same quality has been demonstrated live.

Acceleration uses neighbouring estimated velocities. Values are not joined across period boundaries or long time gaps. Ball height is supplied z data; we do not fit a full 3D flight.

INTERACTION

Player interaction and current carrier

Meaning

Interaction means that a player appears able to act on the ball. An interaction interval is the time from a detected onset until that player can no longer influence it.

1. Start an interval at an acceleration-based touch or a stable slow-proximity onset.

2. Predict where the ball will be one second ahead, using its present estimated velocity.

3. Compare that predicted location with the player's current location. A 4 m reach is the default limit.

4. Assign the carrier only while the ball is within 2.5 m and reachable, or at a detected touch that immediately releases it.

5. Finish an interval when another player interacts, the ball is no longer reachable, tracking breaks, the period changes or the ball goes outside.

Predicted ball location = present ball location + 1 second x ball velocity

Example

A player taps the ball a little ahead while running. The ball can still be within the one-second reach, so the interaction may continue. A fast pass travels far beyond that reach, so the interaction ends.

Measured exact-time agreement

357 of 366 assigned players matched the actor at eligible event timestamps: 97.5% agreement when assigned. Only 366 of 1,668 timestamps received an assignment: 21.9% coverage.

An interaction label is not independently validated sustained control. The algorithm can record a one-touch release without claiming prolonged control.

CONTROL

Likely ball control

Meaning

Likely control is a simple movement-based estimate that the interacting player can manage the ball. It is stronger than a possible touch, but it is not a proven judgement.

1. The current carrier rule must already assign this player.

2. At least 0.15 seconds must have elapsed since the current interaction interval started.

3. The difference between the ball's velocity and the player's velocity must have size at most 5 m/s.

4. If ball height is available, it must be at most 2 m. Missing height does not prove the ball is on the ground.

Example

A player moves at 3 m/s and the ball moves alongside at 3.5 m/s in the same direction. Their relative movement is small, which supports control. A ball racing past the player is less compatible with control.

What accuracy do we have?

No validated true-control accuracy yet. The learned classifier is untrained. The 97.5% carrier/actor agreement is not a validation of this control rule.

What the run produced

The smoothed model labelled about 1,014.58 seconds as likely control. This is an unverified model output, not a measured amount of true control.

The 0.15 s test uses interaction age. It does not prove that every earlier instant in that interval had continuous control. When no reviewed model is loaded, control_probability is blank.

POSSESSION

Team possession: how ownership is followed

Meaning

Team possession is estimated from player interactions. It is not the same as the time that one player physically controls the ball.

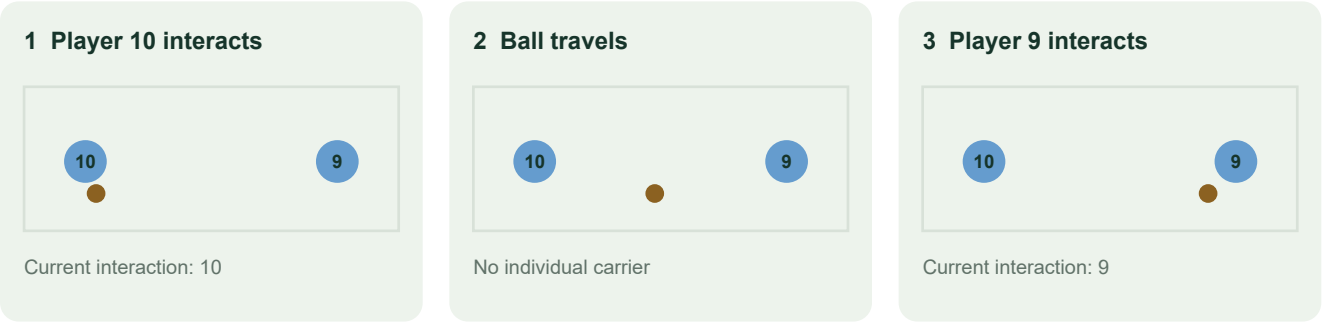
1. When a player interacts, assign their team and remember that player as the latest owner.

2. If no carrier is currently assigned, retain the previous team provisionally for up to five seconds since the last influence.

3. A new player interaction can update the owner and switch the team.

4. Clear current assignments at missing ball positions, out-of-bounds, period boundaries or source gaps over 0.2 s.

5. During a contest, leave carrier and team blank. Keep the previous owner only as history for analysing the challenge.



The in-flight label is provisional

in_flight_provisional means no individual carrier is assigned while the previous team is retained. The current code does not require positive ball speed for this label, so the label alone does not prove the ball is literally flying or moving.

Whole-match possession accuracy is unknown. A continuous, independently validated ball-in-play flag is missing. In-bounds stoppages can still receive an assignment.

POSSESSION

Possession time and percentages

How seconds are counted

Give each frame the time until the next frame. Count it only when the next frame is in the same period and the gap is positive and at most 0.2 s. Do not invent extra time after the final frame.

Team share = team assigned seconds / (home assigned seconds + away assigned seconds)

Measured quantity	Smoothed run
Home / Argentina assigned seconds	1,988.72 s
Away / Saudi Arabia assigned seconds	946.75 s
Both teams assigned	2,935.47 s
All accounted tracking time	6,611.28 s
Share within assigned time	67.7% home / 32.3% away
Assignment coverage	44.4%; the remaining 55.6% is unassigned

All accounted tracking time



44.4% assigned to a team

55.6% unassigned

Within the assigned time only



Do not call this official possession

67.7% and 32.3% describe the assigned subset. They cannot represent complete match possession while 55.6% of accounted time is unassigned and stopped play is not reliably separated.

A separate sparse check finds the team matches the event actor's team at 1,431 of 1,668 checked moments (85.8%). That proxy does not validate the full possession durations.

EVENT RECIPES

Touches: the two ways we find them

In simple words

A touch candidate means the tracking suggests a player interacted with the ball. It does not directly show a foot or body contact.

How we calculate it

1. Route A: find a local peak in the size of ball acceleration, at least 4 m/s². Peaks are separated by roughly 0.15 s, based on the median sample interval.

2. Look within 0.1 s before/after the peak for a nearest player within 2.5 m. Prefer a close approach near the peak time.

3. Route B: the same nearest player stays within 1.8 m, ball speed stays at most 8 m/s, and no contest is detected, for at least 0.1 s.

4. Nearby repeated contacts for the same player are reduced; the stronger peak is retained. An ambiguous touch may still be exported while current carrier/team stay unassigned.

Example

A ball slows sharply near player 9. Route A can detect the change. A very slow ball arriving gently can instead be found by Route B.

How we validate it

Match the correct actor and period to an eligible event record within 0.6 s. Each prediction/reference can be used only once. The file does not label every genuine touch.

Touch actor proxy - measured result

1,553 matched out of 1,668 eligible reference records. There were 2,434 candidates. Reference recall: 93.1%. Precision is not identifiable from this sparse reference list.

Limit to remember

93.1% is recovery of the eligible actor labels within a time window. It is not touch precision and not full-match carrier accuracy.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Kicks: a low-ball contact clue

In simple words

A kick candidate is an acceleration-based touch with a low reported ball position.

How we calculate it

1. Start with a touch found from an acceleration peak. The stable-proximity route by itself does not create a kick.

2. Require a finite reported ball height z of at most 0.5 m.

3. Keep the actor and touch time. A touch can therefore create both a touch record and a kick record.

Example

The ball changes motion near player 10 and its reported height is 0.1 m. We can label a kick candidate, but we have not observed the player's foot.

How we validate it

Compare actor and period within 0.6 s against eligible pass, cross, shot and clearance action records. These are kick-action actor proxies, not a complete labelled inventory of foot contacts.

Kick actor proxy - measured result

690 matched out of 964 eligible reference records. There were 1,683 candidates. Reference recall: 71.6%. Precision is not identifiable from this sparse reference list.

Limit to remember

A low ball does not prove a foot contact. Missing height prevents this particular rule. Headers and ambiguous body contacts are not independently classified.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Completed passes

In simple words

A completed-pass candidate links a ball interaction by one player to a later interaction by a teammate.

How we calculate it

1. Take consecutive different-player contacts in the detected contact sequence. Repeated same-player contacts are not passes.

2. Require the same period and at most five seconds between contacts.

3. Require valid ball positions throughout the path, no source gap over 0.2 s, and all ball coordinates inside the model pitch.

4. Require at least 2 m separation between the ball positions at the two contacts, with both actors on the same team.

5. Use the earlier contact as approximate release time and the later contact as reception time.

Example

Home 10 interacts at 20.0 s. Home 9 interacts 1.2 s later, 8 m away, with a valid tracked path. We record a pass from home:10 to home:9.

How we validate it

Use completed PA/CR reference outcomes C with mapped participants. Require the exact passer, receiver and period. Release time must be within one second. Do not match anywhere along the entire flight.

Completed pass - measured result

592 matched out of 702 eligible reference records. There were 746 candidates. Reference recall: 84.3%. Precision: 79.4%; F1: 81.8%.

Limit to remember

746 candidates include 154 unmatched predictions. The model misses 110 of the 702 references. Short passes, missing paths and false touches still cause errors.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Receptions

In simple words

A reception candidate is the later interaction in a predicted completed pass.

How we calculate it

1. First detect a completed-pass candidate.
2. Create a reception at the receiving player's detected contact time.
3. In this record, actor is the receiver and target stores the original sender.

Example

For a predicted pass home:10 -> home:9, create a reception with actor home:9 and target home:10.

How we validate it

Pass matching checks that the recipient identity agrees with the recorded completed pass. It does not independently verify the exact reception timestamp.

Current result

746 candidate records in the smoothed run. Detection accuracy for this specific subtype has not been measured.

Limit to remember

746 receptions come from the 746 pass candidates. They are related records, not 746 additional unrelated actions. Reception timing accuracy remains unknown.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Unsuccessful passes

In simple words

This candidate means the ball travelled from one player and an opponent subsequently interacted. Intent is uncertain.

How we calculate it

1. Use the same valid-path and maximum five-second checks as the interaction-transfer logic.
2. The next different actor must be on the other team.
3. Require at least 3 m displacement and at least 0.15 s between contacts.
4. Record the earlier actor as the possible unsuccessful passer. Intended recipient is unknown and is left blank.

Example

Home 10 releases the ball. Away 8 interacts with it 10 m away. This can be an unsuccessful pass, but it can also be a clearance or rebound.

How we validate it

Compare the release actor/period within one second with eligible PA/CR records having a non-empty outcome other than C.

Unsuccessful pass - measured result

97 matched out of 193 eligible reference records. There were 270 candidates. Reference recall: 50.3%. Precision: 35.9%; F1: 41.9%.

Limit to remember

The rule cannot reliably tell a deliberate pass from every other long release. It can overlap with an interception or tackle candidate.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Carries: moving with the ball

In simple words

A carry candidate means a player appears to move the ball while retaining control during most of an interaction.

How we calculate it

1. Start from a completed interaction interval for one player.

2. Require duration of at least one second.

3. Require ball start-to-end displacement of at least 3 m.

4. At least 70% of the interval's frames must be labelled likely_control or model_control. This percentage counts frames, not time-weighted seconds.

Example

The player and ball move 5 m over two seconds. If 45 of 60 frames satisfy the control rule, the fraction is 75%, so a carry candidate is created.

How we validate it

Look for the correct player and overlapping time near the provider's BC records, with a 0.6 s allowance. The file contains only nine eligible BC labels.

Carry - measured result

0 matched out of 9 eligible reference records. There were 186 candidates. Reference recall: 0.0%. Precision is not identifiable from this sparse reference list.

Limit to remember

None of the nine BC references matched. Review the meaning and interval boundaries of provider BC labels. This does not establish that all 186 detected carries are false.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Contests: two teams near the ball

In simple words

A contest candidate marks an episode where neither side has a clear distance advantage near the ball.

How we calculate it

1. Find the nearest home player and nearest away player to the ball. These need not be the two nearest players overall.

2. Each must be within 2.5 m of the ball.

3. The difference between those two distances must be less than 0.6 m.

4. Keep episodes lasting at least 0.1 s, splitting at period changes and source gaps over 0.2 s.

5. Current carrier/team remain unassigned. Prior owner is history; an opponent near the ball is recorded as a possible challenger.

Example

A home player is 0.8 m away and an away player is 1.1 m away. Their gap is 0.3 m. If this persists, a contest can be labelled.

How we validate it

Compare the participant pair and period with eligible CH references. Candidate/reference time intervals can overlap, with a 0.6 s tolerance. Reference order of the pair does not matter.

Contest - measured result

31 matched out of 198 eligible reference records. There were 213 candidates. Reference recall: 15.7%. Precision: 14.6%; F1: 15.1%.

Limit to remember

This is a proximity pattern, not proof of a tackle. The measured precision and recall are low, and missing players can hide a real contest.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Who won a contest?

In simple words

We look for the first assigned interaction after the contest. That player is a possible outcome player, not necessarily the final football winner.

1. Remember the prior owner and the likely challenger at the contest.

2. Search forward after the contest for up to 30 source frames and at most one second.

3. Stop searching if the period changes, a source gap exceeds 0.2 s, or ball position becomes unavailable.

4. Use the first non-empty carrier assignment. If none is found, leave the outcome blank.

Example

Home 10 and away 8 contest the ball. The next detected interaction belongs to away 8. Away 8 is the possible winner. A brief touch can still be mistaken for true possession.

How we validate it

First a contest must already match the CH participant pair and timing. Only then compare the proposed winner with challengeWinnerPlayerId.

Matched-subset outcome	Count
Matched contests with winner label	31
Winner assigned	27
Assigned winner correct	13
Assigned winner wrong	14
No winner assigned	4

Agreement when assigned = 13 / 27 = 48.1%

Scope of this result

This small, selected subset does not measure success on every tackle or every contest in the match.

EVENT RECIPES

Tackle, interception and recovery candidates

All three start with the same basic pattern: the next different interacting player belongs to the other team. A valid in-bounds path and the maximum five-second transfer window are required.

Priority	Exact current classification
1. Tackle candidate	A contest occurred anywhere between those contacts and elapsed time is at most 2 s.
2. Interception candidate	If rule 1 did not apply: ball displacement is at least 3 m and elapsed time is at least 0.15 s.
3. Recovery candidate	If neither rule above applied: label the remaining opponent transfer as a recovery candidate.

Example

The opponent interacts immediately after a close contest: tackle candidate. The opponent takes a travelling ball without that nearby-contest pattern: interception candidate.

What is recorded

Time is the new player's interaction time. Actor is the new interacting player; target is the prior actor. The same sequence may also create an unsuccessful-pass candidate.

Smoothed output	Candidates
Tackle	170
Interception	173
Recovery	0

Accuracy is not measured for these subtypes

The classifier is heuristic. It does not prove foot contact, a legal tackle, deliberate interception or sustained winning possession. The intervening contest is not required by this rule to have exactly the same participant pair.

Do not use contest precision as tackle precision. CH participant matching and these subtype rules are different evaluations.

EVENT RECIPES

Retained challenges, turnovers and dribbles

Shared starting point

A detected contest has a prior owner and a subsequent outcome player. Both identities must be available for the retained/turnover label.

Candidate	Calculation	Smoothed count
Challenge retained	The same player is the prior owner and the subsequent interacting player.	18
Challenge turnover	A different player is the subsequent interacting player.	150
Dribble	The same player resumes interaction and the ball moved at least 1 m from contest start to contest end.	6

Example

Home 10 is the prior owner. If home 10 interacts again, label retained. If the contest also involved movement, it can produce a dribble candidate.

Important name limitation

The current challenge_turnover rule means a different player, even if that player is a teammate. The name alone does not prove that the team lost possession.

How we validate it

These specific subtype labels have not been independently evaluated against a confirmed provider subtype dictionary or reviewed clips. Their detection accuracy is unknown.

A nearby opponent and retained interaction do not establish that a defender was beaten by a deliberate take-on.

EVENT RECIPES

Aerial duels and pressure

Aerial duel candidate

Start from a detected contest. If any available reported height within that episode is above 1.5 m, add an aerial-duel candidate. The smoothed run contains 58. We do not observe headers or body contact, and aerial-duel accuracy has not been measured.

Pressure candidate: exact current test

A carrier must be assigned. While that player is interacting, at least one opponent must be within 3 m of the ball for an episode lasting at least 0.2 s. Split the episode at period changes or long source gaps.

Carrier assigned + nearest opponent-to-ball distance <= 3 m + duration >= 0.2 s

Who is the presser?

Record the nearest opponent to the ball at the episode start. That identity is not continually reselected throughout the episode.

Example

Home 10 interacts. Away 8 stays near the ball for half a second. This supports a spatial pressure candidate, but it does not prove effective defensive pressure.

Distance wording matters

The code uses opponent-to-ball distance, not opponent-to-player distance. Because the carrier can be up to 2.5 m from the ball, those distances can differ.

Validation and result

There are 207 pressure candidates. We have not independently measured pressure accuracy. Tracking gaps or absent opponents can underestimate pressure; being nearby is not the same as challenging.

Implementation: [C2]. These categories remain unvalidated subtypes.

GEOMETRY

Which way is the team attacking?

Why we need it

Shot, cross and clearance rules need to know which goal belongs to the attacking team. The model estimates direction from the player positions, without using event outcomes.

1. For each period, take up to its first 150 retained frames.

2. Find the median x position for the home players and for the away players.

3. If the home median is smaller than the away median, assign home attacking direction +1, towards increasing x. Otherwise assign -1.

4. Give away the opposite direction. If either group lacks usable positions, direction remains unavailable.

Forward depth = x position x attacking direction

Example

A team attacking right has direction +1. x = 30 gives forward depth 30 m. A team attacking left has direction -1; x = -30 also gives forward depth 30 m.

Pitch location	Forward-depth test
Attacking third	Greater than +17.5 m
Defending third	Less than -17.5 m
Attacking goal plane	+52.5 m in the team's forward direction

This is an assumption-based estimate

Unusual formations or a partial capture can confuse direction. A missing estimate prevents the direction-dependent candidate rules. It is not independently validated as an attacking-direction model.

EVENT RECIPES

Shots: projected travel towards goal

In simple words

A shot candidate is a fast release whose projected xy path goes near the opposing goalmouth.

How we calculate it

1. Start at an acceleration-based touch with a known attacking direction.

2. Use ball velocity about 0.15 s after the contact, within the same valid tracked segment.

3. Require forward travel, speed at least 8 m/s, and start depth greater than 17.5 m in the attacking direction.

4. Project the straight-line path to the attacking goal plane. Arrival must be more than 0 and at most 4 s ahead.

5. The projected sideways goal location must have absolute y at most 5.66 m. This includes a 2 m margin beyond the 3.66 m half-goal width.

Example

A fast ball released in the attacking third travels towards the goalmouth. It fits the geometry, but a cross can sometimes fit the same shape.

How we validate it

Match SH actor, period and touch/release time within 0.6 s. There are 18 eligible SH references after the label filters, not all 24 raw SH rows.

Shot - measured result

11 matched out of 18 eligible reference records. There were 23 candidates. Reference recall: 61.1%. Precision: 47.8%; F1: 53.7%.

Limit to remember

This rule does not check shot intent, goalkeeper response or the complete curved flight. It does not use a 3D goal-height test to establish a shot.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Crosses: a wide-to-central transfer

In simple words

The current cross candidate is a completed teammate transfer from a wide location into an advanced central area.

How we calculate it

1. A completed-pass candidate must already exist.
2. The release position must be wide: absolute y at least 17 m.
3. The receiving location must be at forward depth at least 30 m.
4. The receiving location must be more central: absolute y less than 20 m.

Example

A wide player sends the ball to a teammate near the central penalty area. This meets the shape of a cross, but intention is not directly observed.

How we validate it

Match the CR actor, period and release time within 0.6 s. The reference set includes 37 eligible cross records.

Cross - measured result

5 matched out of 37 eligible reference records. There were 15 candidates. Reference recall: 13.5%. Precision: 33.3%; F1: 19.2%.

Limit to remember

The rule only considers transfers already detected as completed passes. It therefore misses unsuccessful crosses and does not reproduce a provider's full crossing definition.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Clearances: a pressured defensive release

In simple words

A clearance candidate is a fast forward release from the defending third with an opponent near the ball.

How we calculate it

1. Start at an acceleration-based touch and use ball velocity about 0.15 s later.
2. Require forward travel away from the team's own goal.
3. Require release depth less than -17.5 m and ball speed at least 10 m/s.
4. Require the nearest opponent to be at most 3 m from the ball at release.

Example

A defender under nearby opposition kicks the ball quickly upfield. The same movement might instead be a deliberate long pass.

How we validate it

Match the CL actor, period and release time within 0.6 s. There are 51 eligible clearance references.

Clearance - measured result

1 matched out of 51 eligible reference records. There were 11 candidates. Reference recall: 2.0%. Precision: 9.1%; F1: 3.2%.

Limit to remember

Only 1 of the 51 references matched. This category is currently very weak. Do not use its 11 candidates as a reliable count of actual clearances.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Out of bounds

In simple words

An out candidate starts when the tracked ball centre lies beyond a model pitch boundary.

How we calculate it

1. Require finite ball xy coordinates.

2. Check whether absolute x is greater than 52.5 m or absolute y is greater than 34 m.

3. Group consecutive outside samples into an episode; split at a period change or source gap over 0.2 s.

4. Use the first outside sample as event time and the last as episode end. Current carrier/team are cleared.

Example

A ball centre moves from y = 33.9 m to y = 34.2 m. The model marks an out candidate at the first outside sample.

How we validate it

Compare with OUT timestamps in the same period, within one second. No participant identity is required for this category.

Out - measured result

37 matched out of 108 eligible reference records. There were 44 candidates. Reference recall: 34.3%. Precision: 84.1%; F1: 48.7%.

Limit to remember

Football requires the whole ball to cross. We use the centre and do not identify the last touch. A ball disappearing before the line can cause a missed out event.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Goal-line crossing: not an official goal

In simple words

This is a geometric hint that the ball centre crossed a goal line between the posts at a reported low-enough height.

How we calculate it

1. Find the start of an outside-ball episode. The prior sample must be valid, in the same period and no more than 0.2 s earlier.

2. Require absolute previous x <= 52.5 and absolute next x > 52.5.

3. Interpolate sideways y where the xy segment meets the x goal plane.

4. Require absolute crossing y <= 3.66 m, and finite reported height at the next sample <= 2.44 m.

Example

The xy segment crosses x = 52.5 m at y = 1 m, with reported next-frame z = 0.5 m. It meets the geometric candidate rule.

How we validate it

No reliable allowed-goal validation is performed. The event file can include outcomes whose official status requires referee/VAR interpretation.

Current result

3 candidate records in the smoothed run. Detection accuracy for this specific subtype has not been measured.

Limit to remember

The test uses ball centre and next-frame reported height, not full 3D whole-ball crossing. Offside, fouls and disallowance are not resolved. The three candidates must not be displayed as an official score.

Implementation: [C2]. Validation: [C3]. Counts/results: [B1].

EVENT RECIPES

Restarts: the shared stationary-ball rule

Meaning

A restart candidate looks like a ball being released after remaining roughly stationary. There is no confirmed whistle or continuous stopped-play flag.

1. Start from an acceleration-based touch.

2. Look back approximately one second within the same period. At least three prior samples must be available.

3. Require finite ball positions in the inspected section.

4. Before the release instant, the ball positions must stay within 0.5 m of the lookback section's first position.

5. Use release location to choose a possible subtype. The next page gives the exact order.

Example

The ball stays near the touchline and then accelerates near a player. This can suggest a throw-in restart. A player who paused with the ball can also trigger the general pattern.

What is not confirmed

A stationary ball does not establish that play stopped. The model does not independently reconstruct all fouls, whistles, last touches or legally awarded restarts.

How we validate it

Restart subtypes are not independently validated. A documented provider subtype mapping and reviewed stoppage/restart intervals would be needed for dependable precision and recall.

Smoothed run: 49 generic stationary restarts and two throw-in candidates. Other implemented restart rules produced no candidates in this run.

EVENT RECIPES

Every restart subtype and its priority

The program checks these location rules in order. The first matching location decides the subtype. The shared stationary-ball rule on the previous page must already pass.

Order / candidate	Location test	Smoothed count
1. Kickoff	Distance from pitch centre <= 2 m.	0
2. Corner	Absolute x >= 50 m AND absolute y >= 30 m.	0
3. Throw-in	Absolute y >= 32 m.	2
4. Goal kick	Absolute x >= 45 m AND absolute y <= 12 m.	0
5. Stationary restart	The previous four location tests did not match.	49

Why order matters

A corner is close to a touchline too. Checking the corner rule before throw-in avoids labelling every corner-location release as a throw-in.

What zero means

Zero candidates means this rule did not detect that subtype. It does not prove no such restart occurred. A first kickoff at the capture start may lack the required pre-release stationary history.

Accuracy status

No validated accuracy is available for kickoff, corner, throw-in, goal-kick or generic stationary-restart subtypes. Generic stationary restart is not a confirmed free kick.

UNKNOWN STATES

Why the answer can be unknown

Unknown means the evidence did not satisfy an assignment rule. It does not mean the ball had no owner in the real match.

Reason	What the model does	What would help
Missing / long invisible ball	Clears the current assignment.	Better ball tracking or a reviewed trajectory.
Ball travelling away	Can leave the carrier blank and retain a team provisionally.	The next observed interaction.
Two opposing players close	Marks a contest and leaves carrier/team blank.	Clear post-contest interaction and video review.
No valid contact onset	Does not simply give the ball to whoever is nearest.	A clearer acceleration change or stable slow proximity.
Outside pitch / period break	Resets interaction and possession history.	A new valid interaction in play.
Invalid jump / large time gap	Does not bridge unreliable movement.	A continuous reliable source segment.

The earlier 971 unassigned labels

That number belongs to the old conservative smoothed baseline: 697 of 1,668 event timestamps received a carrier; 971 did not. Its 41.8% was coverage of checked event timestamps, not a measured fraction of every tracking frame.

Coverage has more than one denominator

The current research-style run has 21.9% exact-event carrier coverage, 44.4% time-weighted team coverage, and 97.5% actor agreement when a carrier was assigned. These numbers must always keep their labels.

VALIDATION

Validation: what is the correct answer?

The available reference

The separate events JSON has 2,099 records. It gives selected football events, participants, outcomes and times. It is a useful reference for those actions, but it is not a frame-by-frame video annotation of every touch or every moment of control.

1. Calculate all tracking-based candidates first. Do not use event names, event IDs or possession payloads to decide them.

2. Build a player-ID mapping from event roster snapshots to home/away plus jersey. Exclude ambiguous identity mappings.

3. Ignore rows marked nonEvent = true or videoMissing = true.

4. Choose the event categories and participant fields needed for each validation question.

5. Compare only the same period. Align source video seconds, not a rounded match clock.

6. Match each candidate to at most one reference and each reference to at most one candidate. Export the decisions for inspection.

Independent inputs, limited truth

The predictor ignores embedded labels. However, we do not know whether the provider's smoothed coordinates involved annotation-assisted corrections. This capture is not an independently collected, held-out test set.

Names may be added from events after prediction. A friendly player name in the app does not mean an event label was used to predict possession.

VALIDATION

Exactly which event labels we use

Reference code / field	Used to check
PA: passerPlayerId	Pass sender; PA outcome C plus receiver identifies a completed pass.
CR: crosserPlayerId	Cross actor; CR outcome C plus receiver also enters completed-pass references.
SH: shooterPlayerId	Shot actor.
CL: clearerPlayerId	Clearance actor.
BC: ballCarrierPlayerId	Sparse carry actor; not an exhaustive carry inventory.
TC: touchPlayerId	Touch actor.
RE: rebounderPlayerId	Rebound actor.
IT: gameEvents.playerId	Interception actor used in the generic actor/touch proxy.
CH: carrier + challenger	Contest pair; fallback fields homeDuelPlayerId and awayDuelPlayerId. Optional challengeWinnerPlayerId checks the winner.
gameEventType OUT	Out-of-bounds reference; participant identity not required.

Touch and kick proxies

All eligible actor categories above, except CH, form the touch-actor reference list. PA, CR, SH and CL form the kick-actor proxy. An actor label is evidence of an action; it is not a complete list of physical contacts.

Reference filters change counts

There are 1,668 eligible generic actor labels. The old actor helper excludes 428 unsupported, flagged or ambiguous rows and three unmapped rows from 2,099 input events. Each specialized event metric builds its own eligible list.

Completed pass requires both mapped participants. Unsuccessful PA/CR requires a nonempty outcome other than C; it checks the mapped sender, without guessing an intended receiver. Implementation: [C3], [C6].

VALIDATION

How we align and match events

Two different checks

Exact-frame carrier checking asks who was assigned at a reference timestamp. Event matching asks whether a predicted action occurred near a reference action. These use different time allowances.

Check	Identity and time rule
Exact-frame carrier	Nearest source frame in the same period, at most 0.067 s away; then compare carrier to actor.
Completed pass	Same sender AND receiver, same period, release times within 1 s.
Unsuccessful pass	Same sender and period; release times within 1 s.
Touch / kick / shot / cross / clearance / carry	Same actor and period; tolerance 0.6 s.
Contest	Same unordered pair and period; intervals can overlap or be at most 0.6 s apart.
Out of bounds	Same period and time within 1 s; no actor required.

One-to-one matching

First group candidates and references by period and required identity. Find the largest possible number of valid one-to-one matches; among equivalent-size solutions, prefer smaller total timing error. The implementation uses an assignment solver.

Interval error = maximum of (prediction start - reference end, reference start - prediction end, 0)

Overlapping intervals have zero interval error. For completed passes, unsuccessful passes and crosses, prediction end is set to release time for validation so the whole ball flight cannot make release matching artificially easy.

METRICS

Precision, recall and F1: a simple example

Example using the actual completed passes

The model proposed 746 completed passes. The filtered events file contains 702 completed-pass references. There were 592 valid one-to-one matches with the correct sender, receiver, period and timing.

Count	Meaning	Value
TP: true positive	A matched prediction.	592
FP: unmatched prediction	Candidate not matched to this reference list.	746 - 592 = 154
FN: missed reference	Eligible reference not matched.	702 - 592 = 110

Precision = TP / predictions = 592 / 746 = 79.4%

Recall = TP / references = 592 / 702 = 84.3%

F1 = 2 x TP / (predictions + references) = 1,184 / 1,448 = 81.8%

Say it in everyday language

About 79 out of 100 proposed completed passes match this events file. About 84 out of 100 eligible completed passes in the events file are recovered.

Why not one overall accuracy?

Most moments are not a shot, pass or tackle. Counting those easy negative moments would hide missed events. For sparse event detection we report precision, recall and F1; we do not invent true-negative counts.

RESULTS

All measured event results: smoothed run

Every row uses the same prediction-before-validation process, but its own eligible reference labels and timing rule. These are results on match 3816 only.

Event	Pred.	Refs	Match	Prec.	Recall	F1
Completed passes	746	702	592	79.4%	84.3%	81.8%
Unsuccessful passes	270	193	97	35.9%	50.3%	41.9%
Shots	23	18	11	47.8%	61.1%	53.7%
Crosses	15	37	5	33.3%	13.5%	19.2%
Clearances	11	51	1	9.1%	2.0%	3.2%
Contests	213	198	31	14.6%	15.7%	15.1%
Out of bounds	44	108	37	84.1%	34.3%	48.7%

How to interpret the weaker rows

A detected shot, clearance or contest is still a review candidate. The current rules do not reliably reproduce every provider definition. Clearance and cross performance is particularly weak.

Why pass results are stronger

The pass rule benefits from observing two mapped contacts and the ball moving between them. Detecting intent such as a clearance or cross from geometry alone is harder.

Precision is relative to eligible references

For the categories treated as exhaustive, unmatched predictions count against precision. A missing, differently defined or mistimed provider label can still cause a disagreement. These scores measure agreement under this evaluation protocol.

Detailed matched rows, unmatched counts, mean timing error and per-period metrics are in event_matches.csv and report.json. Source: [B1].

RESULTS

Touches, kicks and carries: special limits

Proxy / sparse category	Predictions	References	Matched	Ref. recall
Touch actor proxy	2434	1668	1553	93.1%
Kick actor proxy	1683	964	690	71.6%
Sparse carry labels	186	9	0	0.0%

Why touch precision is blank

The event file does not list every physical touch. A candidate with no matching event may be a real but unlisted touch. Therefore we cannot count every unmatched touch as false. Precision and F1 are deliberately not reported.

What 93.1% means

A predicted touch with the correct actor was found near 1,553 of 1,668 eligible actor references. It does not mean that 93.1% of all proposed touches are correct, or that ball control is 93.1% accurate.

What 71.6% means

690 of 964 PA/CR/SH/CL actor references had a matching low-height kick candidate. The missing height or low-ball restrictions can exclude real actions.

What the carry result means

No candidate matched the nine eligible BC references using this actor/time rule. Recall is 0% for that very small list. Because the list is sparse, candidate carry precision is not identifiable.

No unmeasured score is filled with a guess

Not measured is different from 0%. Zero recall here is an actual result on nine labels. Unknown precision means the reference cannot answer that question.

RESULTS

Carrier agreement and coverage, fully explained

Exact reference-time outcome	Count
Eligible actor labels	1,668
Assigned carrier	366
Assigned and correct actor	357
Assigned but different actor	9
No assigned carrier	1,302

Agreement when assigned = 357 / 366 = 97.5%

Assignment coverage = 366 / 1,668 = 21.9%

Correct out of all checked labels = 357 / 1,668 = 21.4%

The trade-off

The model is conservative at exact event instants. It often gives no carrier. A high agreement among its assigned cases must therefore be read together with low coverage.

Other diagnostics

Simply choosing the nearest tracked player matches 93.4% of these sparse actors. Searching for the correct carrier anywhere in a +/-0.25 s window finds one for 92.2% of labels. The window search can use later frames, so it is a timing diagnostic, not exact-time accuracy.

Team and control are separate targets

The actor-team proxy matches at 1,431 of 1,668 moments: 85.8%. It does not validate every possession second. None of these scores independently establishes sustained ball control.

RESULTS

Contest winners and everything unvalidated

Contest winner subset

Among the matched contests, 31 have a known reference winner. The model assigns a winner to 27. Of those 27, 13 agree and 14 disagree; four of the 31 remain unassigned.

Winner agreement when assigned = 13 / 27 = 48.1%

This is already conditional on a contest being detected, matched, and having a reference winner. It is not accuracy across every challenge in the match.

Output	What accuracy can we claim now?
Continuous true ball control	Not measured; no independently reviewed dense labels.
Full-match team possession	Not measured; no verified continuous ball-in-play and possession truth.
Reception timing	Not separately validated; completed-pass matching checks the release pair.
Tackle / interception / recovery subtype	Not separately validated; contest score does not establish subtype correctness.
Pressure, aerial duel, retained challenge, turnover, dribble	Not independently validated.
Restart subtypes and official goals	Not independently validated; goal-line crossing is geometry only.

Software tests are not football accuracy

Tests show that rules and the app behave as implemented on test cases. They cannot replace comparison with independently annotated match video.

COMPARISONS

Raw versus smoothed: same capture

The raw and provider-smoothed groups are alternative coordinate sources in the same tracking file. Both then pass through our processing. This comparison does not add a second independent match.

Measure	Raw	Smoothed
Unique frames	198,142	198,142
Interaction intervals	3,712	1,708
Touch proxy reference recall	66.0%	93.1%
Kick proxy reference recall	54.6%	71.6%
Completed-pass precision	47.7%	79.4%
Completed-pass recall	48.4%	84.3%
Completed-pass F1	48.1%	81.8%
Exact actor agreement when assigned	78.1%	97.5%
Exact actor assignment coverage	19.1%	21.9%
Team time assignment coverage	49.0%	44.4%

More coverage is not automatically better

Raw tracking assigns more team time here but has poorer actor agreement and pass scores. Noise can create extra contacts and intervals. A system should report both coverage and correctness.

Do not compare this to a new match

The paper and other datasets use different labels and conditions. These figures only compare the saved runs on capture 3816. Provider smoothing provenance is not documented.

Sources: [B1], [B2]. The JSON reports retain all raw and smoothed category metrics.

COMPARISONS

Why earlier numbers were different

Two implementations exist in this workspace. The old baseline used a more conservative rule set. The current demo uses the research-inspired processing explained in this guide.

Question	Old smoothed baseline	Current smoothed research run
Completed-pass candidates	220	746
Completed-pass matches	207 of 702 references	592 of 702 references
Completed-pass precision	94.1%	79.4%
Completed-pass recall	29.5%	84.3%
Completed-pass F1	44.9%	81.8%
Exact actor assigned	697 of 1,668	366 of 1,668
Exact actor correct / assigned	695 / 697 = 99.7%	357 / 366 = 97.5%
Exact actor coverage	41.8%	21.9%
Unassigned actor labels	971	1,302

Why more passes but fewer exact carriers?

The event pipeline can recover brief contact transitions near an event time. The frame carrier still has a strict current-interaction rule. Event matching allows timing tolerance; exact carrier checking uses a narrow frame alignment. They answer different questions.

Keep model and metric names with the score

The old 99.7%, current 97.5%, touch recall 93.1%, pass precision 79.4% and coverage 44.4% must not be mixed into one accuracy claim.

Sources: [B1], [B3].

CONTROL FEATURES

All interaction features: movement

Each interaction interval becomes one row in control_annotations.csv. The smoothed run contains 1,708 rows. These summarize the ball and player during that interval; they are not extra confirmed football events.

Field / unit	Calculation in simple words
duration_s / seconds	Time at the interval end boundary minus time at the start.
mean_ball_speed / m/s	Average size of the ball's xy velocity over the interval.
var_ball_speed / (m/s)^2	Average squared difference between each ball speed and the mean speed.
mean_ball_acceleration / m/s^2	Average size of the xy acceleration vector.
var_ball_acceleration / (m/s^2)^2	Average squared difference from the mean acceleration size.
mean_ball_distance / metres	Average xy distance between the ball and the interval's player.

A tiny worked example

Suppose ball speeds are 2, 4 and 6 m/s. Mean speed is (2 + 4 + 6) / 3 = 4. Variance is ((2 - 4)^2 + (4 - 4)^2 + (6 - 4)^2) / 3 = 2.67. Higher variance means more changing speed.

Sample average versus elapsed time

These averages give each sampled frame equal weight. They are not weighted by how many seconds each frame represents. Duration is measured in source seconds.

Missing-value conventions

NaN speed/acceleration values are ignored when finite values exist; otherwise their summaries default to 0. If no finite player-ball distance exists, distance defaults to 10 m. These are numeric fallbacks, not observations that the ball was stationary or 10 m away.

CONTROL FEATURES

All interaction features: opponents and quality

Ten opponent-neighbourhood features

For each frame, count opposing players within a circle centred on the BALL. Then average that count over the interval. The opposition is defined relative to the interval player's team.

Exact CSV field family	Radii
opponents_within_0.5m through opponents_within_5.0m	0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5 and 5.0 metres.

Example

At three sampled frames, the number of opponents within 2 m of the ball is 0, 1 and 2. The exported opponents_within_2.0m is 1.0. A decimal such as 0.4 is an average, not part of a player.

Missing opponents

An opponent with unavailable coordinates cannot satisfy the distance test and contributes zero at that frame. The count is not corrected for missing opponent tracking.

Extra diagnostic	How calculated
mean_tracking_quality	At each frame take the smaller of the player and ball quality weights; average these values.
reconstructed_fraction	Number of frames whose ball xy was short-gap reconstructed divided by the number of frames in the interval.

Features do not validate themselves

Movement summaries, neighbourhood counts and quality diagnostics are measurable inputs. We have no independently labelled physical-sensor accuracy for these quantities. Their role is to support review and a future control model.

The optional classifier uses 16 movement/neighbourhood features. These two quality diagnostics are exported for inspection but are not classifier inputs.

CONTROL FEATURES

The annotation sheet: every remaining column

The full annotation CSV has 27 columns: six movement features, ten opponent features, two quality diagnostics, and the nine identity/review columns below.

Column	Meaning / what to enter
action_id	Internal interaction number; join related frame rows.
player	Team-plus-jersey key, such as away:17.
period	Source period for this action.
start_time_s	First interaction source timestamp in seconds.
end_time_s	Interval end boundary in seconds. Feature samples use the half-open start-to-end index range. At the final capture frame the last available timestamp is used.
capture_id	Source tracking filename by default. Preserve a unique, stable capture identity when merging sheets.
reviewed	Blank until manually reviewed; enter yes to include the row in training/evaluation.
label_control	Enter 1 for reviewed control or 0 for reviewed interaction without control.
split	Enter train or test. Keep test examples separate from training examples.

What a reviewer must judge

Watch the complete interval in synchronized video. Decide whether the player could deliberately manage the ball, rather than merely touching it. Do not create a control label just by copying a pass event or the current heuristic.

Review is still required

The exported sheet contains blank review labels. No genuine control training accuracy has been produced from these blank rows.

CONTROL MODEL

The optional learned control model

What is implemented

A small reviewed-label Gaussian naive Bayes classifier can learn whether an interaction interval represents control. It uses the six movement features and ten opponent-count features: 16 inputs in total.

1. Select only rows with reviewed = yes. Labels must be 0 or 1, and split must be train or test.

2. Require at least 20 training examples of each class and five test examples of each class.

3. Require capture_id. In the same capture and period, a training interval and a test interval must not overlap or lie less than one second apart.

4. Require all selected numeric model inputs to be finite.

5. For each class, calculate each feature's mean, variance and the class frequency.

6. Compare how plausible a new interval's features are under the two classes; normalize the two scores to produce the control score.

Decision rule

Score at least 0.5 gives model_control; lower scores give model_interaction_without_control. The interval score is copied to its assigned-carrier frames. It does not newly assign a carrier where the carrier rule is blank.

Current status

This benchmark did not load a trained model. control_probability is blank. Only the heuristic likely_control rule runs. The paper's trained network and its published accuracy are not available as our model.

The optional model assumes features are conditionally independent given the class. Its numeric posterior is not calibrated proof of correctness. Implementation: [C4].

FUTURE VALIDATION

How we would validate true control and possession

Control needs reviewed intervals

Use synchronized video and at least one trained reviewer to label interaction starts, ends and control/non-control. Resolve unclear examples and check agreement between reviewers. This additional annotation has not yet been completed.

1. Keep a final test set separate, ideally using entire unseen matches rather than nearby intervals from the same game.

2. Train and choose any thresholds using only the training/development data.

3. On the frozen test set, report a confusion matrix: true control/non-control versus predicted control/non-control.

4. Report control precision, recall and F1, plus no-answer coverage and interval-boundary errors.

5. For possession, annotate a continuous ball-in-play flag and the actual player/team timeline, including passes, stoppages, contests and unknown video.

6. Compare time-weighted timelines and team shares only on eligible annotated time. Report seconds correct, seconds wrong, unknown seconds and duration errors.

What training already reports

If a reviewed model is trained, it writes test correct/total and a 2 x 2 confusion matrix, with actual class as rows and predicted class as columns. Additional per-class scores can be calculated from those counts.

No future score is promised

Better labels and more matches allow a defensible measurement. They do not guarantee a particular accuracy. Current control and full-match possession remain unverified.

APP GUIDE

The app: loading and running a recording

1. Open the Possession Demo at the local web app, using the possession view.

2. Select a tracking JSONL file. Choose Raw or Smoothed to match the coordinate groups present in your file.

3. Optionally select the matching events JSON. Prediction still works without it, but accuracy comparisons require usable event references.

4. Press the analysis button. Selecting a file alone does not upload or start processing.

5. Watch upload progress and analysis stages. You can cancel the current job; only one analysis runs at a time.

6. When ready, inspect the replay, current state, candidate events, player totals and validation tab. Export the supporting files.

Automatic source hint

The browser checks the beginning of the file, up to 64 KB. If it does not find smoothed groups, it selects Raw. You can still choose the source explicitly; the hint is not a full schema validation.

When validation is unavailable

With no events file, or no usable mapped references, the app shows no inferred accuracy percentages. Add compatible events and run again to obtain comparisons.

Local processing limits

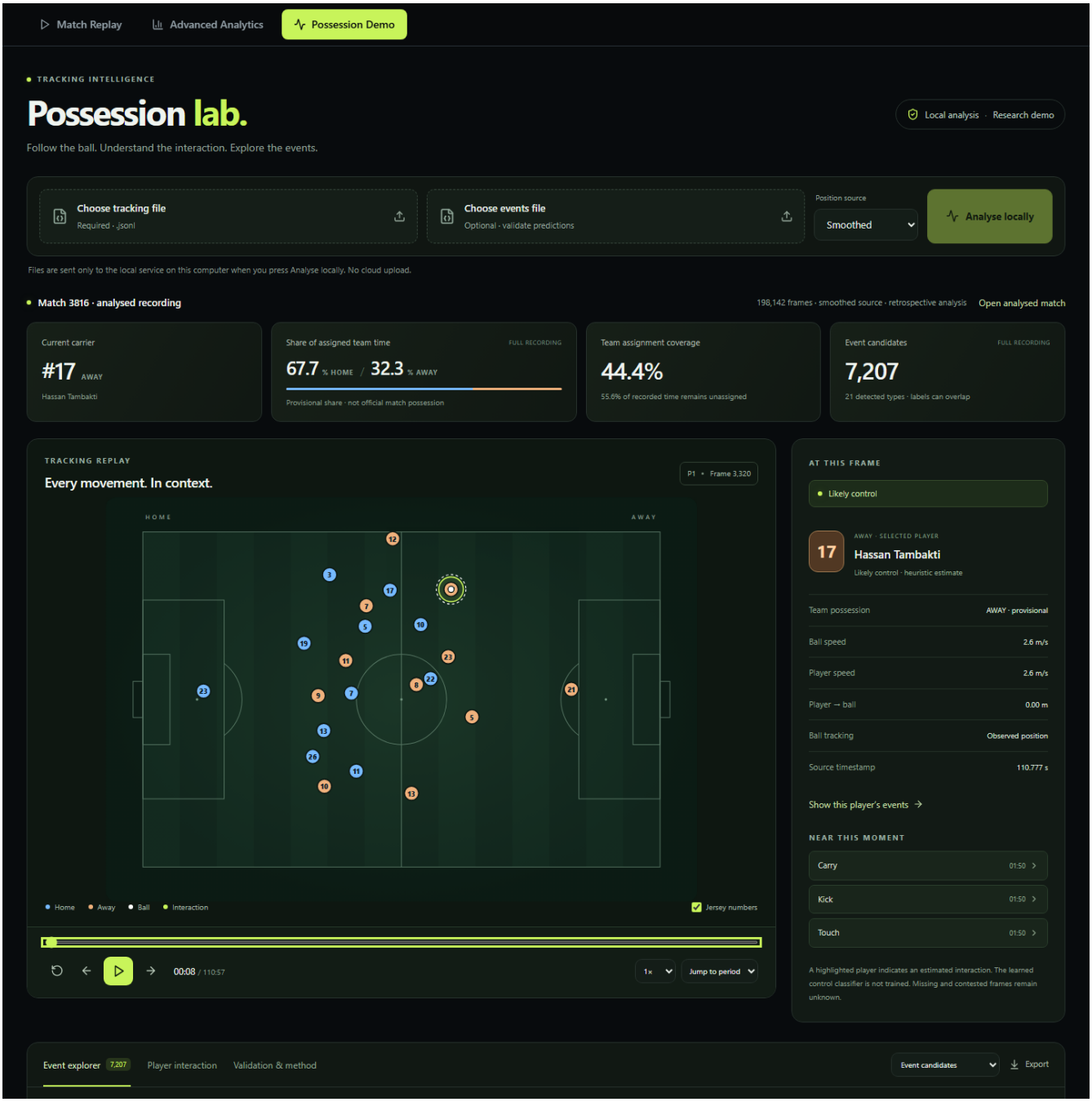
The upload path accepts about 1 GB of tracking data and up to 64 MiB of events data. The background worker has a ten-minute timeout. Unsupported structure, size limits or parsing errors are surfaced as job errors.

This is a local prototype using a Python API and the existing web app. It has not been deployed as a public, multiuser service. Implementation: [C7], [C8].

APP GUIDE

The demo screen: how to read it

A saved screen from the implemented local demo. It illustrates the interface; the benchmark tables in this guide remain the source for measured results.



Pitch: tracked positions and current carrier highlight. Side panel: the selected player and current source moment. Lower tabs: event candidates, player summaries and validation. A selected player is not automatically the carrier.

APP GUIDE

The app: replay, tables and downloads

Feature	What it means
Pitch replay	Draws processed player/ball xy. The carrier highlight follows the inferred current interaction.
Play, pause, seek	Move the source-time playhead; jump five seconds or choose a period.
Speed	0.5x, 1x, 2x, 4x or 8x playback. Requests update about every 100 ms.
Source clock	Uses source video time, not the official match clock. Large source gaps are skipped during playback.
Current player / ball speed	Magnitude of processed xy velocity in m/s; missing values stay unavailable.
Event table	Filter candidate types, click a timestamp to seek, inspect actors and targets. Several labels can describe one action.
Player table	Interaction seconds, detected touches and completed-pass candidate counts for each player.
Validation tab	Per-category matched/reference counts, precision or recall where supported, and explicit unknown measures.
Exports	Download report JSON/Markdown, frame and event CSVs, matches, actor checks, player totals and the review template.

Totals versus current moment

Summary cards and player totals cover the complete processed recording. They are not running totals up to the playhead. The current-state panel follows the selected source moment.

The API returns the nearest processed frame; the replay is a sampled display, not a new 10 Hz prediction model or interpolation of every source frame.

OUTPUT DICTIONARY

Frame export: every field, part 1

frames.csv contains one row per retained source frame. Blank cells mean unavailable or not applicable; do not replace blank carrier values with the nearest player.

Column	Meaning
frame_num	Original source frameNum. Not the internal event array index.
video_time_s	videoTimeMs divided by 1,000.
period	Source period.
carrier	Current inferred interacting player, or blank.
team	Assigned team at this frame, including provisional retention, or blank.
possession_owner	Remembered player attached to the assigned possession history; not necessarily physically touching now.
previous_owner	Owner remembered before this frame's updates, including during a contest; not proof of the current carrier.
status	Main interaction/possession state, explained on the next page.
control_status	unknown, interaction_only or likely_control in the untrained run. An optional model can supply model_control or model_interaction_without_control.
control_probability	Optional reviewed-model score. Blank in the current benchmark.

All player identifiers in these columns use team:jersey. A player name is a display lookup, not the model's persistent identity.

OUTPUT DICTIONARY

Frame export: every field, part 2

Remaining column	Meaning
nearest	Closest finite tracked player to the ball. It is a diagnostic, not automatically the carrier.
ball_x, ball_y	Processed xy ball coordinates in centred metres.
tracking_quality	Ball observation/reconstruction quality weight at this frame; not a correctness probability.
reconstructed_ball	True when the ball xy was short-gap reconstructed.
action_id	Internal interaction-interval identifier when applicable.

status	Plain meaning
interaction	Current physical-influence estimate, without the likely-control label.
likely_control	The heuristic relative-motion, age and height tests passed.
contested	Both teams close with similar distances; no carrier/team assigned.
in_flight_provisional	Team retained from prior influence without a current carrier.
missing_ball	No usable ball position.
loose_ball	No current eligible interaction or retained possession.
out_of_bounds_candidate	Ball centre beyond a model pitch boundary.
outside_period	Outside a positive playing period.

Status and control_status describe related but distinct questions. They should not be interpreted as a universal confidence ranking.

OUTPUT DICTIONARY

Event export: every field

predicted_events.csv is time-ordered. It is the common event list used by the replay and category-count summaries.

Column	Meaning
event_type	Specific candidate label, such as completed_pass_candidate.
period	Source period.
time_s	Candidate start or release time in source seconds.
end_time_s	Episode end or transfer destination time where relevant; a point event can end at its own start.
actor	Player performing the candidate action. For a reception, this is the receiver.
target	Meaning depends on type: receiver, previous opponent, next contest winner or presser. May be blank.
evidence	Short rule/reason label; not a calibrated probability.
frame_index	Zero-based internal frame-array position, supplied on touch records; different from source frame_num.
ambiguous	Touch ambiguity flag when opposing proximity made the situation contested.
strength	Touch peak acceleration magnitude, or 0 for stable proximity; not a probability. Derived kick rows do not copy these three touch-only fields.
challenger	Opposing participant on a contest when available.

Interpret actor and target by event type

For completed passes actor -> target is sender -> receiver. For reception it is receiver -> sender. For contest target is the inferred subsequent winner. For pressure target is the presser selected at episode start.

OUTPUT DICTIONARY

Validation exports: every field

event_matches.csv

Each row records one successful match, not every rejected pairing. Unmatched counts are in report.json; identify unmatched items by subtracting matched indices from their complete lists.

Field	Meaning
category	Evaluation category, e.g. completed_pass.
prediction_index	Zero-based row index in the complete predicted-event list.
event_index	Zero-based position in the original events JSON array.
timing_error_s	Point-time difference or interval separation after the category's matching rule. Overlap can give zero.
winner_matches	True/false only when the matched contest has a mapped reference winner; blank otherwise.

actor_validation.csv

One row per eligible actor-reference check, with these 15 columns:

Columns	Plain meaning
event_index, event_type, event_time_s, period	Locate the original reference and source time.
expected_player, predicted_player, predicted_team, status	Reference actor and the aligned predicted state.
aligned, clock_error_s	Whether a same-period frame was close enough, and its absolute clock difference.
correct_player, correct_team	Exact actor and actor-team agreement.
nearest_player, correct_nearest	Diagnostic comparison with simply choosing the closest player.
found_in_window	Correct carrier appears in the separate +/-0.25 s diagnostic window.

OUTPUT DICTIONARY

Summary, players and trajectory exports

report.json and REPORT.md

The JSON is the detailed machine-readable result. It records method/source, retrospective flag, parameters, input audit, trajectory settings, model-training status, frame/action counts, candidate counts, time summaries, validation, per-period results and limitations. The Markdown report is a readable summary.

players.csv

Three fields: player, name and interaction_seconds. Rows summarize players with interaction time. Name may be blank without a compatible roster. The API can still display every tracked key and compute its touch/pass counts from the candidate list.

trajectories.npz

A compressed numerical cache for the replay. It stores processed motion and quality; it does not contain the event reference labels. It is produced on disk, rather than exposed through the CSV/report download allowlist.

Arrays	Meaning
time, period, frame	Source seconds, period and original frame numbers.
keys	Stable team:jersey list; indexes the player axis.
ball, xy	Ball N x 2 and players N x P x 2 positions. N = frames, P = player keys.
ball_velocity, player_velocity	Estimated xy velocity vectors with corresponding dimensions.
acceleration	Ball xy acceleration vectors.
player_quality, ball_quality	Processing weights per observation.
height, reconstructed_ball	Reported ball z and short-reconstruction mask.

Loading the NPZ uses allow_pickle=False. It is a cache of this run, not a trained model. Implementation: [C5], [C7].

EVENT INVENTORY

Every implemented event label: 1 of 2

All labels below have an implemented rule. Counts come from the current smoothed run. Zero means no candidate was emitted, not that no such event happened.

Exact event_type	Candidate count
touch_candidate	2434
kick_candidate	1683
completed_pass_candidate	746
reception_candidate	746
unsuccessful_pass_candidate	270
carry_candidate	186
contest_candidate	213
tackle_candidate	170
interception_candidate	173
recovery_candidate	0
challenge_retained_candidate	18
challenge_turnover_candidate	150
dribble_candidate	6

Overlapping labels

A single real-world action can produce several event rows. Do not add the categories together to claim a number of unique football actions.

Across all types: 7,207 rows and 21 types with at least one candidate. There are 25 implemented candidate-label types in total.

EVENT INVENTORY

Every implemented event label: 2 of 2

All labels below have an implemented rule. Counts come from the current smoothed run. Zero means no candidate was emitted, not that no such event happened.

Exact event_type	Candidate count
aerial_duel_candidate	58
pressure_candidate	207
shot_candidate	23
cross_candidate	15
clearance_candidate	11
out_candidate	44
goal_line_crossing_candidate	3
kickoff_candidate	0
corner_candidate	0
throw_in_candidate	2
goal_kick_candidate	0
stationary_restart_candidate	49

Overlapping labels

A single real-world action can produce several event rows. Do not add the categories together to claim a number of unique football actions.

Across all types: 7,207 rows and 21 types with at least one candidate. There are 25 implemented candidate-label types in total.

TIME TOTALS

Every frame-state duration in this run

State totals use the same valid next-frame time differences as possession time. Each accounted instant has one main status, so these status totals sum to the accounted tracking time.

Main state	Seconds	Share of accounted time
interaction	185.95	2.8%
in flight provisional	1,734.93	26.2%
likely control	1,014.58	15.3%
contested	85.99	1.3%
out of bounds candidate	13.58	0.2%
missing ball	3,524.79	53.3%
loose ball	51.45	0.8%

Two useful totals

Interaction plus likely control is about 1,200.53 seconds. Adding provisional team-retention time gives about 2,935.47 team-assigned seconds. These are different totals because a team can retain a provisional assignment without a current carrier.

What the large missing-ball total means

About 3,524.79 seconds of accounted time have no usable processed ball xy. That directly limits how much can be inferred. The status total alone does not tell us whether the cause was camera loss, absent source coordinates or quarantine of an invalid transition.

Not an official ball-in-play clock

Accounted time is 6,611.28 seconds of valid adjacent source intervals. It can include stopped play in the recording. Source gaps over 0.2 seconds and period transitions do not add time.

Minor displayed differences are rounding. No extra seconds are appended after the last frame.

PLAYER TOTALS

How every player total is calculated

Interaction seconds

For frames whose carrier equals this player, add eligible next-frame time differences. This includes interaction_only and likely_control frames. Do not assign provisional ball-flight time to the player's physical-interaction total.

Touch count

Count touch_candidate event rows whose actor is this player. A kick may also be a touch, but kick rows are not added again to the touch count.

Completed-pass count

Count completed_pass_candidate rows whose actor is this player. This is predicted completed passes, not independently verified true passes for the player.

Name, jersey and side

The team:jersey key is the identity. The API splits the key to display the side and jersey. A compatible events roster can supply a name after prediction; otherwise the key is displayed.

Tiny example	Result
Player is the carrier in three 0.033 s intervals	About 0.099 interaction seconds.
Three touch rows and two kick rows for the player	Three touches, not five.
Four completed-pass rows and three reception rows	Four completed passes made; receptions are separate.

Accuracy status

Player totals inherit the carrier/event errors and the tracking gaps. No separate per-player ground-truth duration accuracy has been measured. The global pass score should not be assumed identical for every player.

COMPARISONS

All raw-run event metrics

This table completes the alternative-source comparison. It is useful for understanding the impact of coordinate quality, not for selecting a score after looking at labels.

Event	Pred.	Refs	Match	Prec.	Recall	F1
Completed passes	713	702	340	47.7%	48.4%	48.1%
Unsuccessful passes	409	193	48	11.7%	24.9%	15.9%
Shots	16	18	4	25.0%	22.2%	23.5%
Crosses	7	37	0	0.0%	0.0%	0.0%
Clearances	21	51	1	4.8%	2.0%	2.8%
Contests	284	198	42	14.8%	21.2%	17.4%
Out of bounds	163	108	55	33.7%	50.9%	40.6%

Sparse category	Predictions	Matches / refs	Ref. recall
Touch proxy	4588	1101 / 1668	66.0%
Kick proxy	3106	526 / 964	54.6%
Carry	41	0 / 9	0.0%

Sparse-category precision and F1 remain unavailable. Raw-run winner agreement is 18 correct out of 36 assigned, or 50.0%, among 42 matched contests with winner labels.

REPRODUCIBILITY

Reproduce a run and inspect a disagreement

Start the app from the workspace

In a PowerShell terminal in E:/2awashet, run npm run demo:possession. The launcher starts or reuses the local web service on port 5173 and the API on port 8010.

Run the same analysis directly

Use the project Python environment and the research CLI. The following is one command; the events argument is optional.

```
.venv/Scripts/python.exe scripts/research_possession.py --tracking data/tracking_3816.jsonl --events data/events_3816.json --source smoothed --output output/my-review
```

Investigate a specific pass

Open predicted_events.csv and find its row. In event_matches.csv look for that row's prediction_index. A match supplies the original event_index and timing error. Open that original event record, then inspect synchronized replay/video around both times.

For an unmatched candidate

Check identity mapping, participant pair, period, release time, ball gaps and coordinate jumps. Then inspect video to decide whether the candidate was wrong, the reference was incomplete, or the definitions differed. Do not silently change a label just to improve the score.

Train a reviewed model only after annotation

Run .venv/Scripts/python.exe scripts/research_control.py --annotations path/to/reviewed.csv --output output/control-model.json. To apply it later, add --control-model output/control-model.json to a research run.

This guide describes saved benchmark runs and a code inspection. Creating the PDF did not rerun the full match or retrain the model.

TECHNICAL APPENDIX

Technical appendix: the movement filter

You can skip this page on a first reading. It records the precise smoothing assumptions so the simpler explanations can be audited.

State and forward prediction

Treat x and y separately. Each coordinate has state [position, velocity]. For time step dt, predicted position = position + dt x velocity; predicted velocity = velocity. The transition matrix is [[1, dt], [0, 1]].

Process and observation uncertainty

Use continuous white-acceleration process covariance Q = q x [[dt^3/3, dt^2/2], [dt^2/2, dt]]. q is 16 for players and 400 for the ball. These are covariance-model settings, not measured maximum accelerations.

Observation standard deviation = 0.3 m / max(quality weight, 0.15); observation variance R = standard deviation squared

Update and reset

Compare the observed position with predicted position. Blend using the Kalman gain: predicted uncertainty / (predicted uncertainty + observation variance). Update position and velocity from this residual. At a segment start, initialize position to the observation, velocity to zero, and covariance to diagonal [R, 25].

Backward improvement

The Rauch-Tung-Striebel pass adjusts each forward state from the following smoothed state and the transition covariance. It runs only across finite observations in the same period with 0 < dt <= 0.2 s. Missing coordinate channels reset rather than being smoothed through a long hole.

Height and acceleration

Height is the supplied z coordinate, not smoothed by this xy filter. Ball acceleration at i uses (velocity at i+1 - velocity at i-1) / (time at i+1 - time at i-1). It is unavailable across a period break or a two-step gap above 0.4 s.

Implementation: [C1].

TECHNICAL APPENDIX

Technical appendix: contacts and classifier

Contact peak selection

The acceleration route uses scipy.signal.find_peaks on acceleration magnitude, height at least 4 m/s^2 and sample separation round(0.15 / median source dt). For a peak, search same-period frames within +/-0.1 s for an actor within 2.5 m.

> Contact-selection score = nearest-player distance + 2 x absolute time offset from the peak

Choose the lowest score. The numeric time penalty corresponds to 2 metres per second of offset. Consecutive same-player contacts within 0.15 s in the same period are deduplicated by strength. The slow-proximity fallback uses <=1.8 m, ball speed <=8 m/s and an uncontested span of at least 0.1 s.

Classifier variance protection

For each feature, the minimum Gaussian variance is max(0.01 x overall training-feature variance, 0.0001). Each class variance is raised to at least that floor. Class priors are the training fractions.

How the optional score is formed

For each of the two classes, add the log prior and the log Gaussian likelihood for all 16 features. Subtract the largest log score before exponentiating, then normalize the two positive weights. The control-class weight is the output score.

Why a constant interval score

The model classifies the whole interaction, so all assigned-carrier frames in that interval receive the same model probability and model control status. The untrained heuristic can vary frame by frame instead.

Chosen defaults, not fitted certainty

Distance, speed and duration cutoffs in this guide are the current implementation defaults. They have not been optimized on a separate development set or proven optimal for this provider.

Implementation: [C2], [C4].

RESEARCH CONTEXT

Research inspiration and our differences

The research idea, simply

Link and Hoernig separate possession from the shorter intervals when a player can influence the ball, and classify control within those intervals. They smooth tracking, use distance and ball acceleration to locate interaction starts, and predict ball movement to find interaction ends. [R1]

How their control model was checked

Their control classifier was a trained Bayesian network using reviewed video labels and movement/opposition features. Their published results belong to their data and evaluation, not to our demo. [R1]

What we implemented

We use a related sequence of movement preparation, contact candidates, influence intervals, a separate control estimate and provisional possession. The event recipes, missing-data policy, height checks and thresholds are our explicit adaptation.

Difference in this demo	Practical consequence
No verified continuous ball-in-play flag	Cannot produce trustworthy official full-match possession.
Untrained reviewed-label control model	Current control is a heuristic without measured true-control accuracy.
Optional Gaussian naive Bayes, not their K2 network	A future trained model would still be a different classifier.
One capture and sparse provider events	Our current scores have a narrower validation scope.

We did not reproduce the complete paper

The research supports the general approach. It does not supply this app with a trained model or transfer its accuracy to our tracking file.

SOURCES

Evidence and implementation references

The calculation rules in this guide were checked against these workspace sources. Source paths are relative to E:/2awashet.

ID	File and role
C1	scripts/research_tracking.py - loading, quality, duplicate handling, interpolation and RTS movement preparation.
C2	scripts/research_events.py - contact, interaction, heuristic control, possession, feature extraction and all event recipes.
C3	scripts/research_validation.py - category references, one-to-one matching and event scores.
C4	scripts/research_control.py - reviewed-label model, training safeguards, probability and application.
C5	scripts/research_possession.py - pipeline, summaries, CLI and artifact exports.
C6	scripts/tracking_possession.py - original baseline, roster and exact-frame actor validation helper.
C7	apps/api/possession_demo.py - upload job, cancellation, playback data and export routes.
C8	apps/web/src/components/possession/ - replay, controls, player/event tables and validation UI.
C9	docs/POSSESSION_DEMO.md and docs/RESEARCH_POSSESSION.md - scope, operation and previous verification record.

If brief interface help text differs from these precise rules, the implementation is the basis of this guide. In particular, pressure and the neighbourhood-count features use opponent-to-BALL distance.

SOURCES

Benchmark files and reading reference

ID	Evidence
B1	output/research-possession-3816-smoothed/report.json and companion CSV exports - main results in this guide.
B2	output/research-possession-3816-raw/report.json - alternative raw-source comparison.
B3	output/possession-3816-smoothed/report.json - earlier conservative baseline; source of the old 41.8% coverage.
Input	data/tracking_3816.jsonl and data/events_3816.json - the evaluated capture and separate provider events.
Screen	output/possession-demo/demo-desktop.png - saved implemented-demo screenshot.

R1. Daniel Link and Martin Hoernig (2017). Individual ball possession in soccer. PLOS ONE 12(7): e0179953. DOI: 10.1371/journal.pone.0179953.

Read the open-access paper: pmc.ncbi.nlm.nih.gov/articles/PMC5503225/

Verification record for the implementation

The existing project record reports 49 Python regressions, 12 isolated Chromium browser regressions and 133 current-workspace JavaScript unit tests passing, plus TypeScript, lint and production build checks. An existing opt-in performance test was not run by the ordinary suite.

The broad root test command also discovers old unrelated checkouts under tmp, where three copied suites have workspace-alias errors. The existing Advanced Analytics build chunk warning remains. These are recorded software checks, not fresh checks run for this PDF and not football-event accuracy. [C9]

What this document represents

A plain-language explanation of the current implementation and saved benchmark evidence, prepared on 2 November 2026. It does not claim a trained control model, complete possession truth or validated official-event adjudication.